WELCOME!

Online Training Course

Course 2: Foundations in Mobile Phone Data (MPD) for Policy: Modern Data Workflows

Course 2: Foundations in Mobile Phone Data (MPD) for Policy: Modern Data Workflows

Overview

Eight online training sessions (live webinars):

- Designing mobile phone data for policy pipelines
- Privacy-by-design
- **Input data quality assurance and cleaning**
- Continuity models
- PoIs/Home Locations / Usual Environment
- Calculating "foundational" / "core" aggregates
- Quality Assessment of Estimates for Decision-Making
- From Aggregates to Estimates - population inference from mobile phone data and standard data sources

Five self-paced topics (on OLC):

- Recap of fundamentals of mobile phone data, data pipelines, data products, data types
- "Introduction to the course and examples use cases: data pipeline and stages
- Skill sets for each stage. Objectives."
- Mobile Phone Data: datasets, fields, types (links with 1.5)
- Reference and Complementary data etc.
- Considerations for multi-operators data pipelines

GDF GLOBAL DATA FACILITY GOBIERNO DE ESPAÑA MINISTERIO DE ECONOMÍA, COMERCIO Y EMPRESA THE WORLD BANK IBRD · IDA | WORLD BANK GROUP ITU FLOWMINDER.ORG NOMMON positium

Housekeeping information

- Delivery timeslot for online training sessions (live webinars):
  - Every Wednesday until Wednesday 10 December
  - From **1pm-2pm UTC**

- The slides and a session recording will be uploaded to the Open Learning Campus after delivery, for your reference
- Additional resources, quizzes and/or other follow-up will also be made available after sessions
- Try to keep Q&A until the end

Input Data Quality Assurance and Cleaning

Module 2 - Session 2.3.4

Session Overview

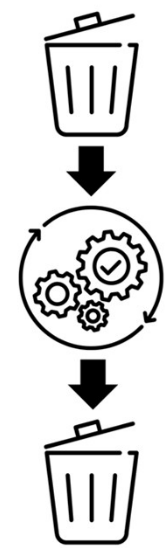
- Recap **why data quality is important** and what **input Mobile Phone Data (MPD) quality assessment** is.
- Introduction to input MPD quality assessment **dimensions**.
- Introducing **indicators**, how to calculate them, and how to read **quality reports**.
- Introducing the concept of **filtering data** and 4 example filters.
- Coding exercise: input data quality indicators and reporting.

Why Is Data Quality Important?

The Critical Role of High-Quality MPD

- High-quality MNO data is a **fundamental requirement** for producing **reliable**, **accurate**, and **trustworthy statistics**.
- It ensures the integrity and usability of the processed data, which is **essential for informed decision-making** and for avoiding inaccurate or misleading statistical outcomes.
- Quality is one of the most important factors for **maintaining public trust,** along with data privacy.

Garbage in, garbage out (GIGO)



Common Issues in Raw Data

Common Record-Level Issues

```
ms_id|timestamp|cell_id
34f9834rhj384j9384|2019-01-01 12:00:01 ^M10052
34f9834rhj384j9384|2019-01-01 12:15:13|47461
34f9834rhj384j9384|2019-01-01 18:30:24|10052
34f9834rhj384j9384|2019-01-01 18:30:24|10052
34f9834rhj384j9384|2019-01-01 20:04:12|30461
34f9834rhj384j9384|2019-01-01 20:05:55
34f9834rhj384j9384|2019-01-02 00:30:28|20490
34f9834rhj384j9384|02/01/2019 01:23:00|30461
34f9834rhj384j9384|2019-01-02 01:25:10|30461
34f9834rhj384j9384|2055-01-03 12:00:00|20490
NULL|2019-01-01 12:15:13|20490
```

hidden characters

duplicates

missing delimiters

inconsistency in data types

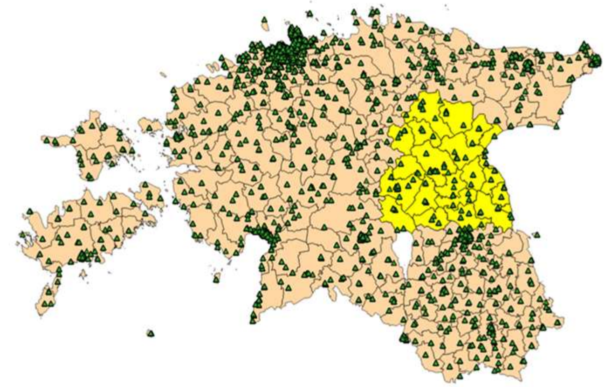
impossible timestamps

missing values

etc.

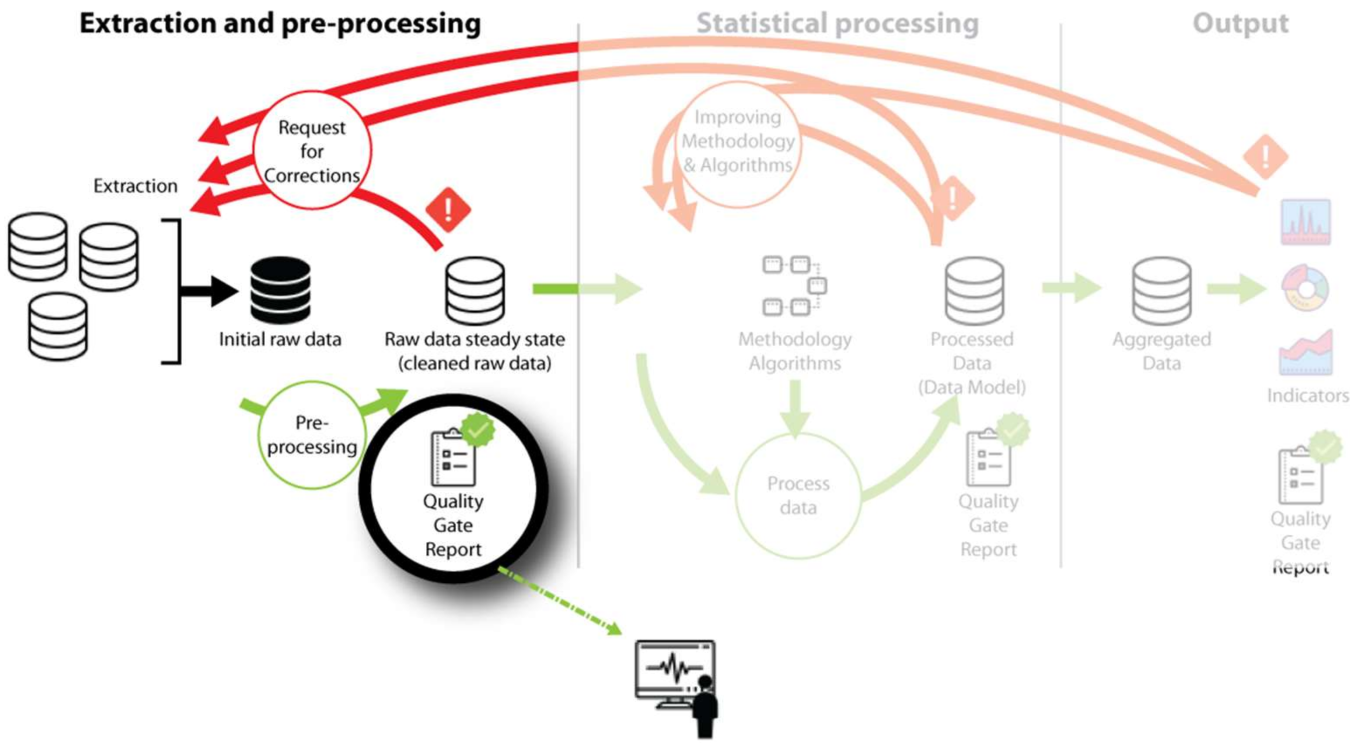
Common Issues in Raw Data

- Data gaps
- Time zone issues
- Wrong cell coordinates or attributes
- Changes in continuity of the IDs
- IoT devices
- Human activities that are not part of the target population
- …



MPD Quality Assessment Framework

Quality Assurance Framework: Quality Gate 1 – Raw Data



Data Quality Dimensions

Data quality assessment is a **multi-dimensional process**.

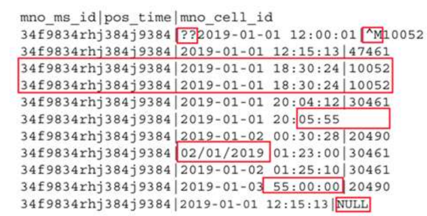
COMPLEXITY

- **File size**: checks if the data amount matches expectations.
- **Attributes**: a syntactic check that verifies fields are present and correctly formatted according to a predefined schema.
- **Time**: a semantic check that looks for logical consistency in timestamps.
- **Space**: a semantic check that assesses geographical validity.
- **Context**: the most advanced dimension, which uses complex rules to determine if the data makes logical sense in the real world.

Types of Data Quality Checks

Syntactic checks

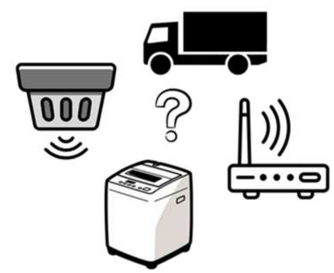
- **Focus:** Data format & structure
- **Core question:** *Does the data look right?*
- **Purpose:** Ensures the data is in the correct "shape" to be processed and won't cause system errors.
- **Example:** Is the timestamp in the correct format? Are all the fields present?



```
mno_ms_id|pos_time|mno_cell_id
34f9834rhj384j9384|??2019-01-01 12:00:01|^M10052
34f9834rhj384j9384|2019-01-01 12:15:13|47461
34f9834rhj384j9384|2019-01-01 18:30:24|10052
34f9834rhj384j9384|2019-01-01 18:30:24|10052
34f9834rhj384j9384|2019-01-01 20:04:12|30461
34f9834rhj384j9384|2019-01-01 20:05:55
34f9834rhj384j9384|2019-01-02 00:30:28|20490
34f9834rhj384j9384|02/01/2019 01:23:00|30461
34f9834rhj384j9384|2019-01-02 01:25:10|30461
34f9834rhj384j9384|2019-01-03 55:00:00|20490
34f9834rhj384j9384|2019-01-01 12:15:13|NULL
```

Semantic checks

- **Focus:** Data meaning & logic
- **Core question:** *Does the data make sense?*
- **Purpose:** Confirms the data reflects real-world events, ensuring accuracy and consistency.
- **Example:** Can a single device be in two different locations at the same time? Is a SIM card associated with a human or a machine?



Quality Assessment Framework for MPD

- Quality assurance is a **foundational aspect** of our entire data processing methodology, not just an add-on.
- Quality-focused components:
  - **Cleaning** components: automatically fix minor data issues.
  - **Quality warning** components: flag significant issues for review.
  - **Quality reporting** components: system measures data quality and provides reports.
- Key dimensions: **file size, attributes, time, space,** and **context**.
- Both **syntactic** and **semantic** indicators are used to provide a complete picture of data quality.

MPD Quality Assessment Indicators

Syntactic Data Quality Indicators

Indicator	Description
Invalid record count	How many records in the data file are not syntactically valid.
Invalid value count	How many records in the individual columns are not syntactically valid.
Duplicate row count	Count of identical records.
Cell occupancy	Assesses the relationship between cells and CDR files.

Invalid Records/Values

How many records in the data file are not syntactically valid.

How many records in the individual columns are not syntactically valid.

CDR:

```
ms_id|timestamp|cell_id
34f9834rhj384j9384|2019-01-01 12:00:01|10052
34f9834rhj384j9384|2019-01-01 12:15:13|NULL
34f9834rhj384j9384|55:00:00|NULL
```

```
Invalid values in records: 2
----------------------------
Invalid values ms_id: 0
Invalid values timestamp: 1
Invalid values cell_id: 2
```

Duplicates

Count of identical records.

CDR

```
ms_id|timestamp|cell_id
34f9834rhj384j9384|2019-01-01 12:00:01|92224
34f9834rhj384j9384|2019-01-01 12:15:13|10052
34f9834rhj384j9384|2019-01-01 12:15:13|10052
```

```
Duplicate records: 2
```

Cell Occupancy

Cells that lack corresponding records in CDR datasets.

CDRs that reference cells missing from the cells data (cells without location data).

CDR

```
ms_id|timestamp|cell_id
34f9834rhj384j9384|2019-01-01 12:00:01|cell_11
34f9834rhj384j9384|2019-01-01 12:15:13|cell_22
34f9834rhj384j9384|2019-01-01 12:17:26|cell_33
```

CELLS

```
cell_id|lat|lon
cell_11|26.73081|58.37060
cell_22|26.65342|57.43520
cell_44|26.23986|58.76549
```

```
Cell occupancy
---------------------------
cell_id-s with no records: cell_44
cell_id-s with no location data: cell_33
```

Syntactic Data Quality Indicators Reports Example

file_name	indicator_name	rows_affected	prc_of_total_rows	pass_fail
cdr_20250914.csv	invalid_row_count	1500	1.5%	pass
cdr_20250914.csv	invalid_value_count_ms_id	367	0.4%	pass
cdr_20250914.csv	invalid_value_count_timestamp	0	0%	pass
cdr_20250914.csv	invalid_value_count_cell_id	1206	1.2%	pass

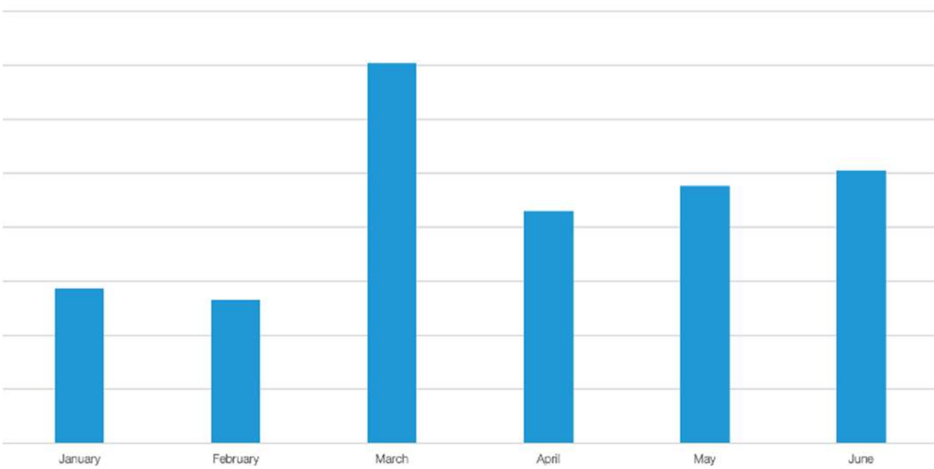
[INFO] [2025-09-14 10:00:00] Data Quality Check: cdr_20250914.csv - Invalid Row Check. Status: **Passed (1.5% < 5% threshold)**.

Semantic Data Quality Indicators

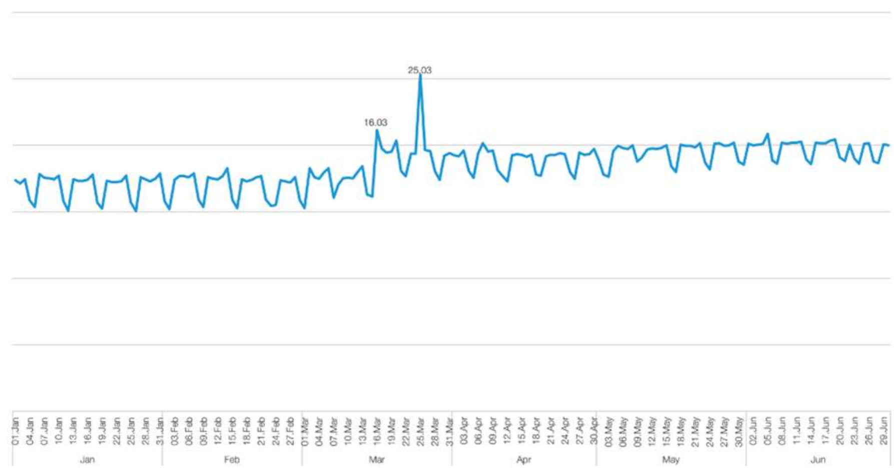
Indicator name	Description
Record/unique subscriber/cell count (hourly, daily, weekly, monthly)	Identifying logical temporal patterns (diurnal, daily, weekly, seasonal).
Spatial distribution of cells/records/users	Analyzing cell information for anomalies/inconsistencies.
Subscriber presence	Confirming persistent and expected MNO subscriber continuity.
Time between subsequent records	Distribution of number of record pairs and gap in hour.
Record count per subscriber per day	Distribution of records per subscriber.
Same time different location duplicates	Same subscriber different locations at the exact same timestamp.

Unique Subscriber Count Per Period

Unique subscriber count per month.

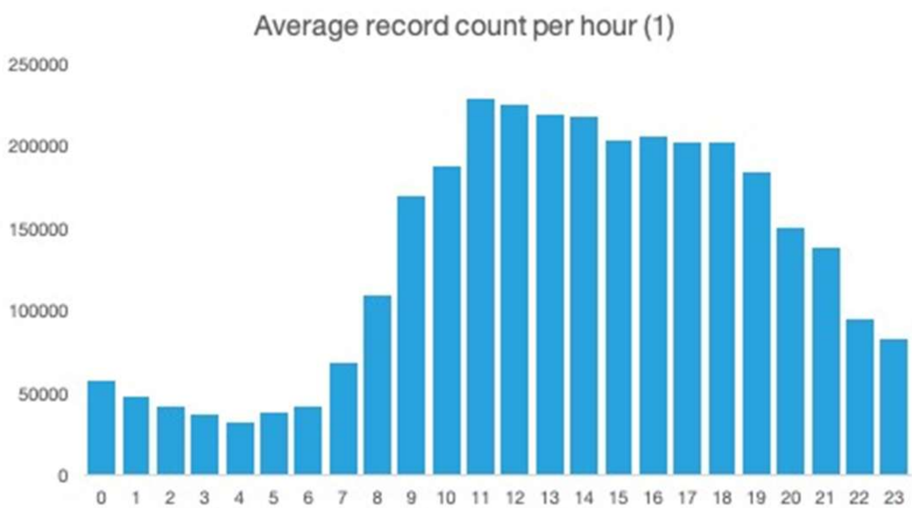


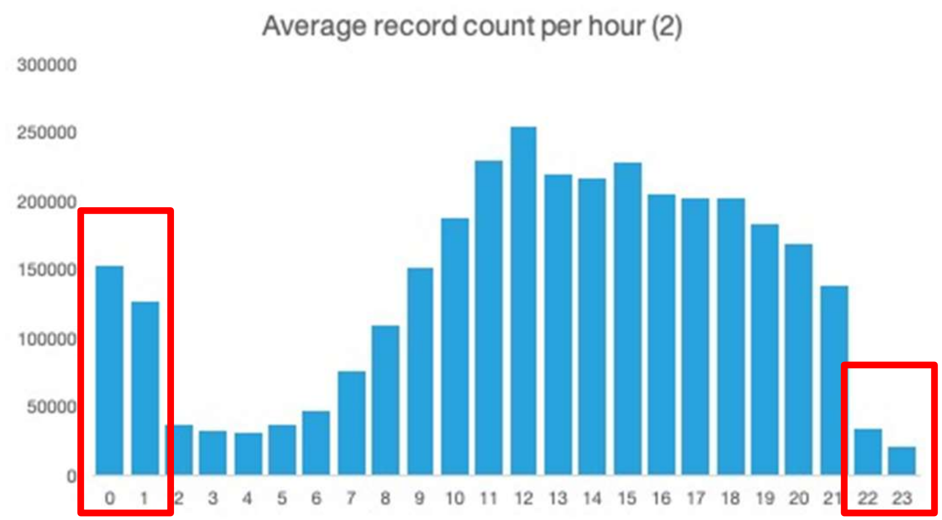
Unique subscriber count per day.



Diurnal Distribution of Records

Average number of records per hour (0-23).

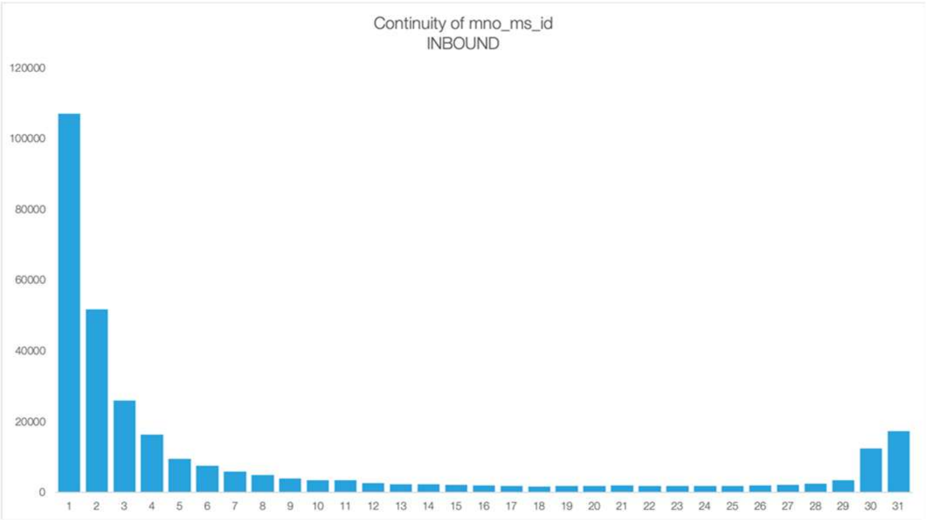


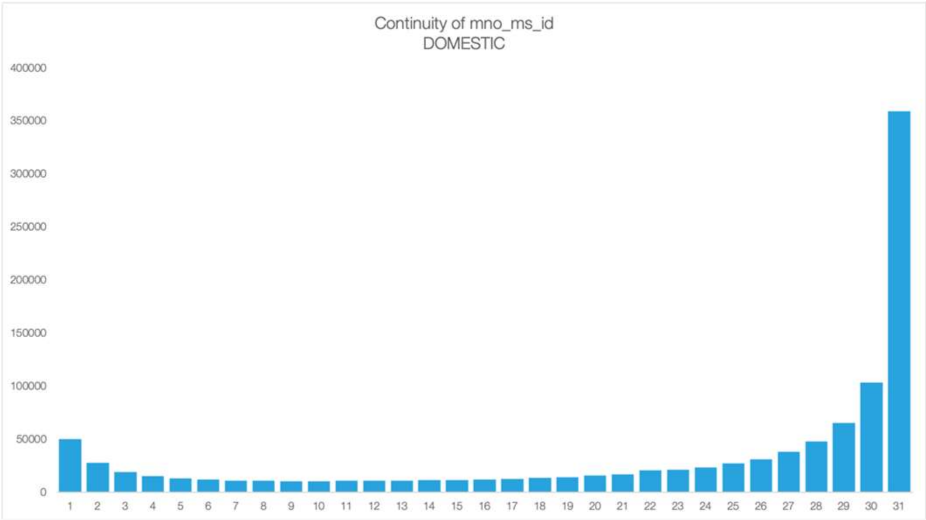


GDF GLOBAL DATA FACILITY · THE WORLD BANK · ITU · FLOWMINDER.ORG · NOMMON · positium

Subscriber Presence

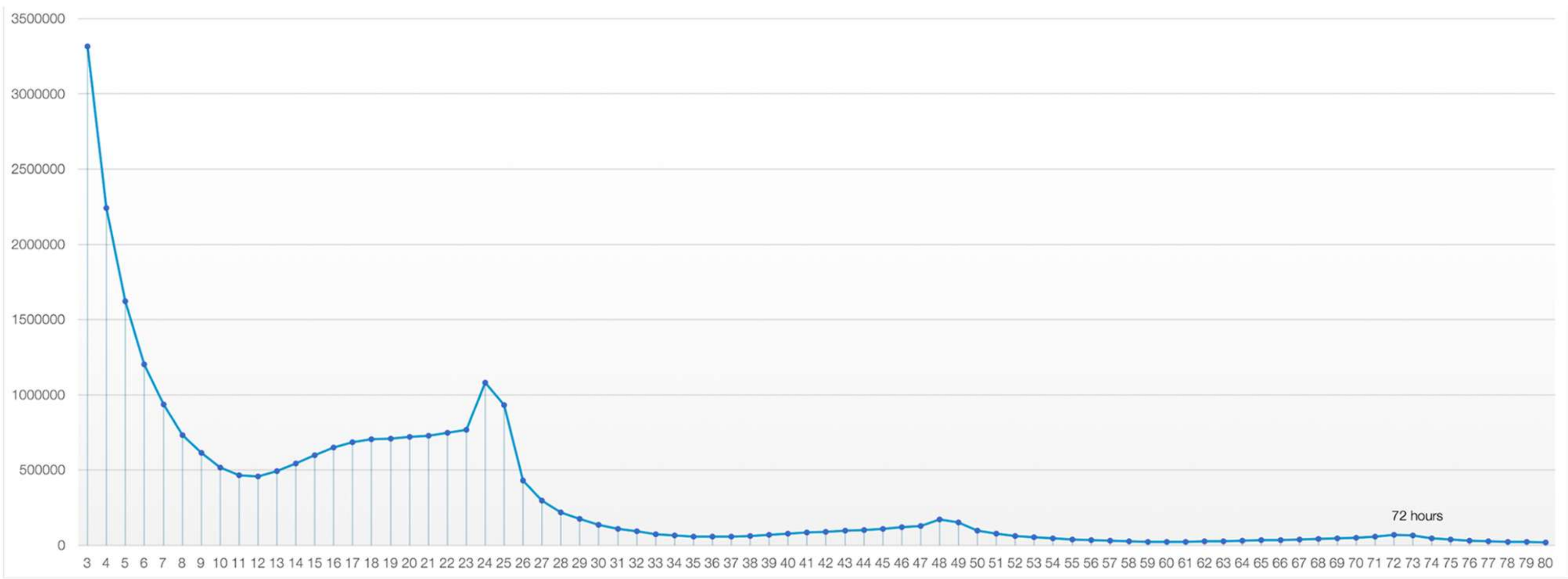
In how many days, subscribers are present out of all days in the period.





Time Between Subsequent Records

Time gap between subsequent records.



Spatial Distribution of Cells

- Verify that all cells are **located inside country borders**.
- Analyze the distribution of cells to identify any missing regions or **significant gaps in network coverage**.
- **Check for expected patterns**, such as a higher density of cells in urban areas or near major highways, and a lower density in rural areas.

Semantic Data Quality Indicators Reports Example

date	record_count	stdev_from_mean	pass_fail
…	…	…	…
2025-09-14	985,500	-0.72	pass
2025-09-15	1,020,000	1	pass
2025-09-16	250,000	-37.5	fail
2025-09-17	245,000	-37.75	fail
…	…	…	…

```
[INFO] [2025-09-14 12:00:00] Data Quality Check: Daily record count. Status:
Failed (2025-09-16, 2025-09-17) - Detected a critical drop of >3.0 standard
deviations from the mean.
```

Automation vs. Non-Automation In Data Cleaning

Automated data cleaning

- Automated cleaning is **essential for handling large volumes of data**.
- Pre-defined syntactic and semantic rules to **automatically detect and fix errors**.

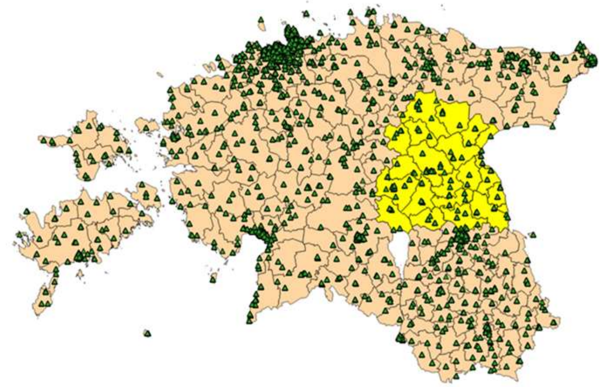
  For example, a system can automatically drop duplicate rows or correct an invalid date format without human oversight.

- The system can be programmed to **automatically terminate the process** if quality metrics fall below a set threshold.

Non-automated data cleaning

Non-automated cleaning is typically used for complex, ambiguous, or rare issues that are difficult to define with a simple rule.

Automation vs. Non-Automation In Data Cleaning

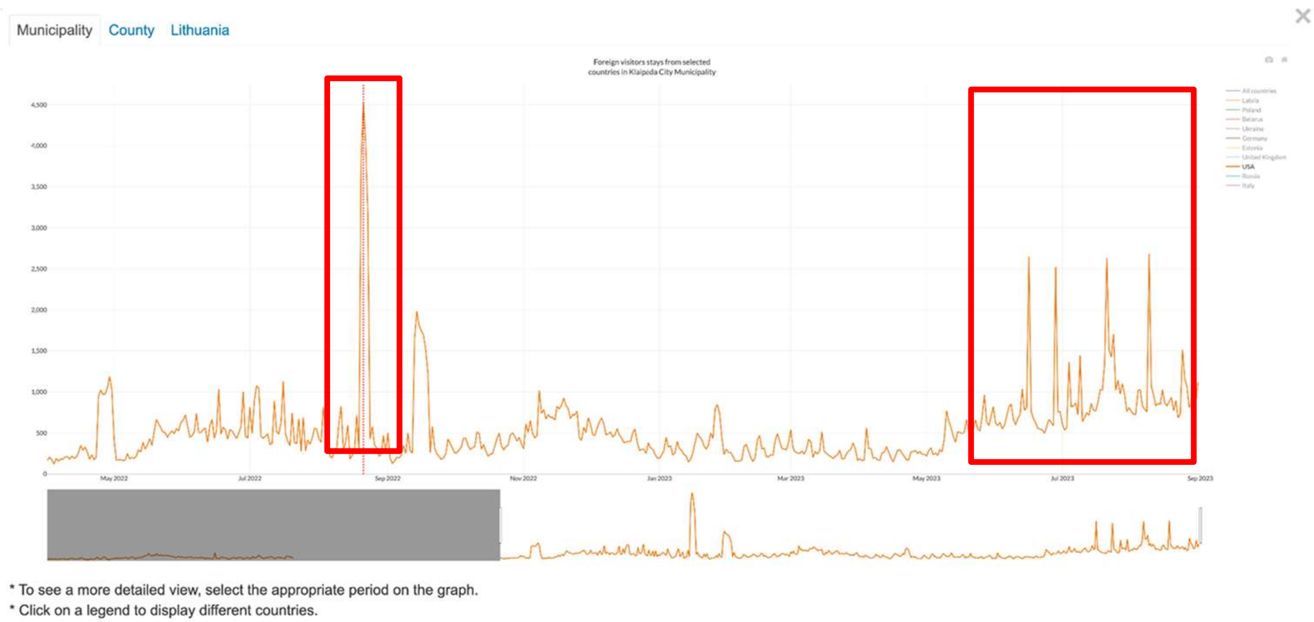


A severe **storm-caused power outage took out the mobile communications network** of a whole region for more than two days in 2013.

The effect was a lack of mobile communication for the population and subsequent data gaps in the mobile phone data.

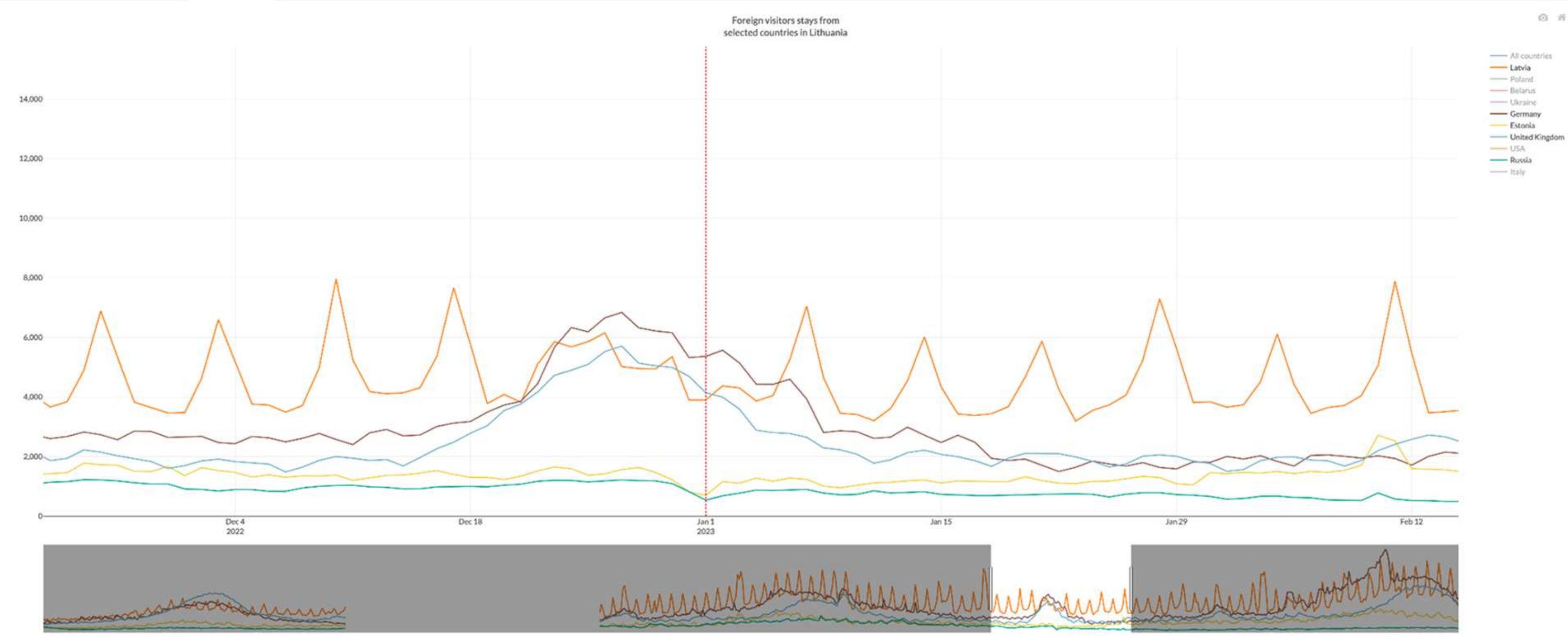
Automation vs. Non-Automation In Data Cleaning

Peaks in Klaipeda City Municipality in August 2022.



Automation vs. Non-Automation In Data Cleaning

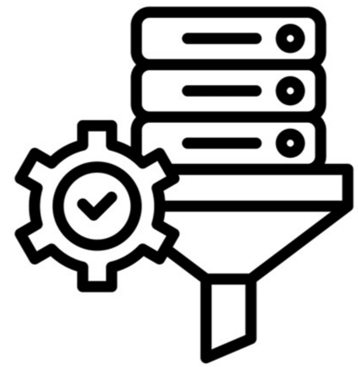
Municipality County Lithuania



* To see a more detailed view, select the appropriate period on the graph.
* Click on a legend to display different countries.

Filtering Out-of-Scope Devices

Filtering Out-of-Scope Devices



The Challenge:

- We want our dataset to only contain records from the population of interest.
- Raw mobile phone data often contains records from sources that are not aligned with the intended methodology of statistical production.
    - Internet of Things (IoT) devices (e.g. security cameras, GPS devices in delivery trucks)
    - human activities that are not part of the target population (e.g., people in planes or ships, accidental roamers in border areas).

The Solution:

- Implementing a series of **smart data filters**.
- These filters analyze the data for specific **spatio-temporal patterns** and contextual clues.

Data Filtering Examples: Removing IoT Devices

Many automated devices use cellular SIM cards to operate. Their data appears in mobile phone records, but it's **not related to human behavior**.

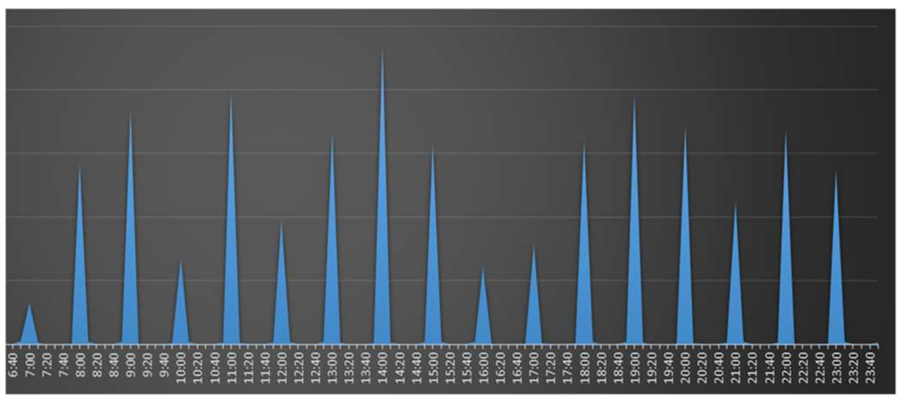
Examples of IoT devices found in Mobile Phone Data:

- **Vehicle & asset trackers:** GPS devices that transmit location data for fleet management and logistics.
- **Smart meters & PoS terminals:** stationary devices that send utility data or process payments.
- **Security & remote systems:** cameras, alarms, and sensors that send alerts or status updates.
- etc.

Data Filtering Examples: Removing IoT Devices

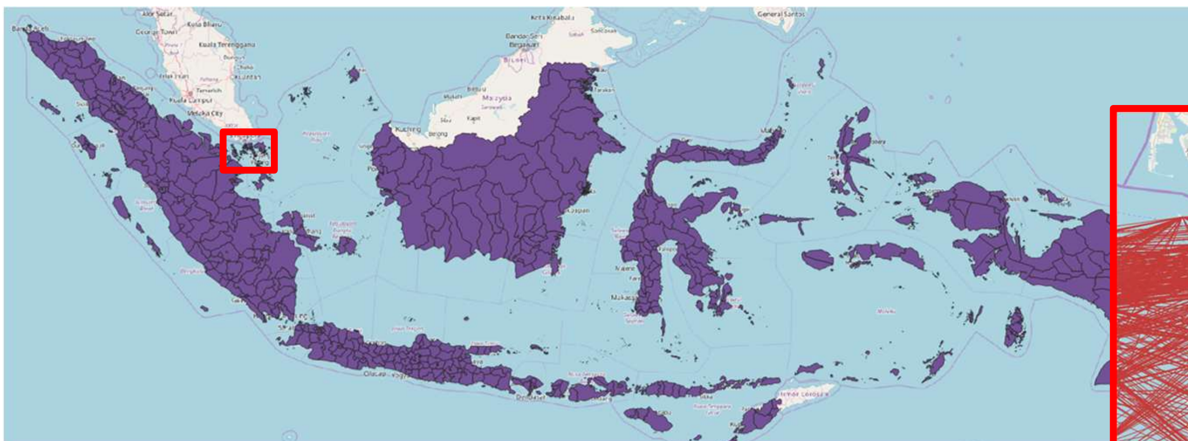
Detecting IoT devices is based on identifying non-human, systematic patterns.

- **Regularity:** data is sent at fixed, consistent intervals (e.g., every 5 minutes).
- **High record count:** a device may generate a very high volume of data compared to a human.
- **Stationary behavior:** the device shows no movement over an extended period.



Data Filtering Examples: Removing "Fast Flyers"

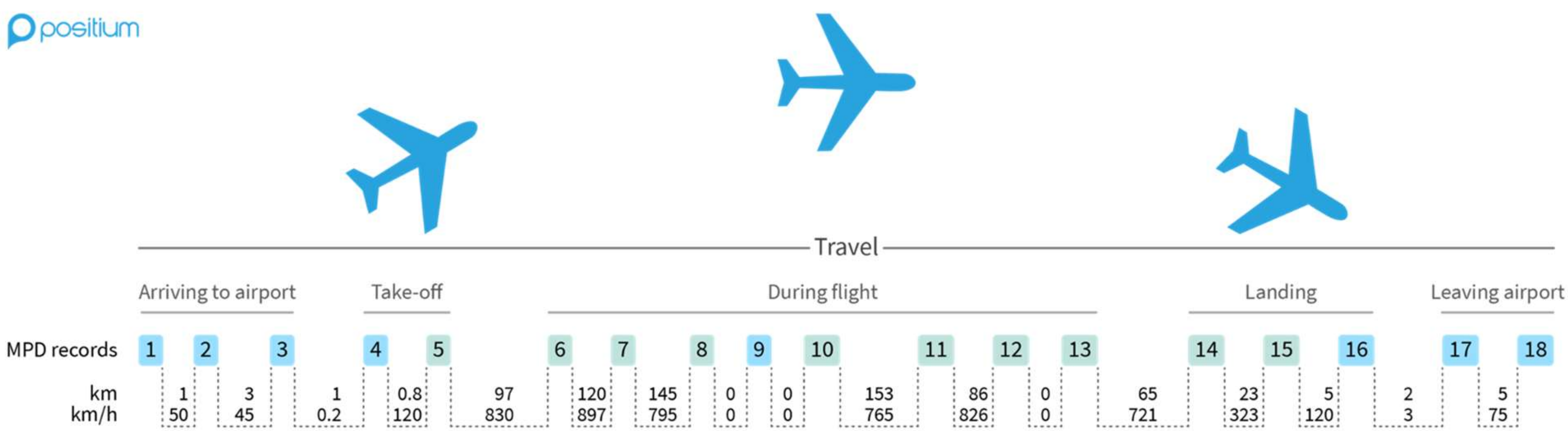
Despite regulations, people keep their **phones turned on during takeoff, landing, and flight**. While in the air, their phone will attempt to connect to distant cell towers on the ground.



High-speed travel patterns most pronounced near Singapore.

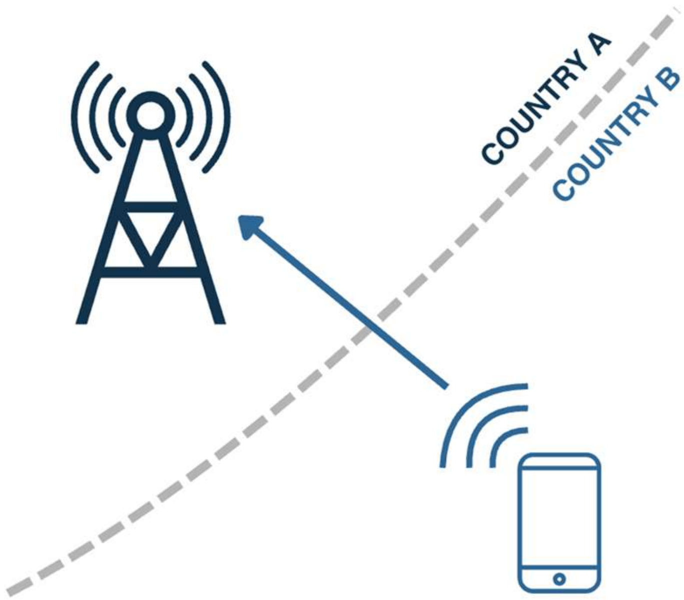
Data Filtering Examples: Removing "Fast Flyers"

Analyze the **speed** of the device and the **unusual change in location over a very short period of time**.



Data Filtering Examples: Removing Accidental Roamers

- People near a border can have their **phones accidentally connect to a neighboring country's cell tower**.
- This results in records that are technically valid, but are contextually inaccurate for population analysis.
- These records can artificially inflate the number of "visitors" to a country.

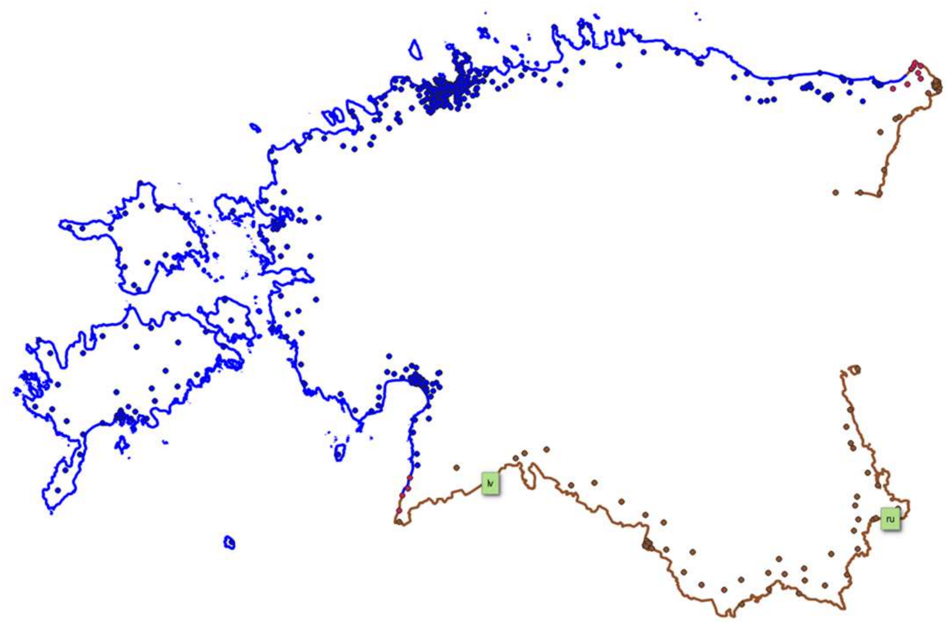


GDF GLOBAL DATA FACILITY · GOBIERNO DE ESPAÑA · MINISTERIO DE ECONOMÍA, COMERCIO Y EMPRESA · THE WORLD BANK IBRD • IDA | WORLD BANK GROUP · ITU · FLOWMINDER.ORG · NOMMON · positium

Data Filtering Examples: Removing Accidental Roamers

- We first **identify border cells** as those located within a predefined distance from the national border.

- We then apply a filter to identify users who show **presence in these border cells** for only a very **limited amount of time**.



Examples of typical filters: Removing Ship Activities (misidentified as on-ground)

Seamen tend to cross the country's seas and create a lot of noise in the data. This noise is unusable and should be filtered.

Cargo and tankers in Estonia

Cargo and tankers in Indonesia

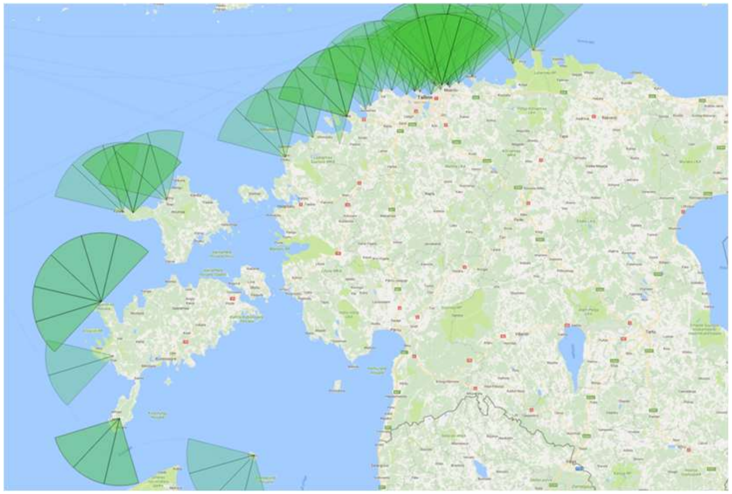
Examples of typical filters: removing ship activities (misidentified as on-ground)

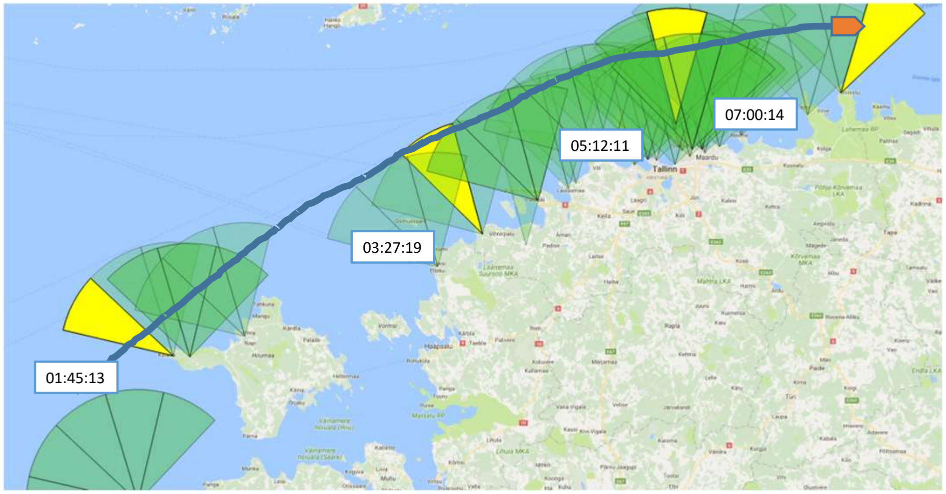
Identify the coastal antennae directed to the sea:

- Based on data from MNOs (coverage areas)
- Buffer zones
- Based on assumable sea traffic

Eliminating **“seamen” roaming**:

- Candidates: **presence in sea-cells**
- Avg speed of movement & std. dev. ~ **speed of marine vessel**





Cleaning Antennae Data

- Besides cleaning the mobile phone data records themselves, it is also crucial to **detect and fix errors in the antennae data**.
- Inaccurate antennae coordinates can lead to significant errors in the final statistical product.

Cleaning Antennae Data

We analyze user **movement patterns** to detect inconsistencies that reveal probable cell location errors.

We **look for cells that often have the next data record in a cell far away** and then mark that cell as 'potentially faulty'.

Example:
Cell A has many data records adjacent to Cell B and C's data records. CEP recognizes this and marks Cell A as 'potentially faulty'.

Coding Exercise

Key Takeaways

Key Takeaways

- What are the most common **MPD quality issues** and their impact?
- What are the key **indicators** of high-quality MPD?
- What techniques detect and remove out-of-scope devices?
- To what extent can input data quality assessment be **automated**?
- How much effort is needed for input MPD quality assurance?

Any Questions?

Thank you!

The GDF-MPD window is a multi-partner window funded by the Spanish Government, Ministry of Economy, Commerce and Business. The GDF-MPD program is jointly managed by DEC (DECDG, DIME), Poverty and Digital Development Global Practices in technical partnership with the International Telecommunications Union (ITU).